Algorithm 2: Power of two-sample test

Input : Two independent random samples $\{X_1, \ldots, X_n\}$ and $\{Y_1, \ldots, Y_n\}$, a threshold ϵ, number of replications r, level of significance α

Output: Estimated power of the test

for $i = 1$ **to** r **do**

  Scale data points of both the samples by their Euclidean norms ;

  Compute T_{n_1,n_2} *for the scaled data under* H_0^{**} *and* H_1^{**}

end

Estimate the critical value c by $\hat{c} = \left(1 - \frac{\alpha}{2}\right)^{th}$ quantile of r values of T_{n_1,n_2} under H_0^{**}

Estimated power $= \frac{\sum_{i=1}^{r} 1_{\{T'_{i,n_1,n_2} > \hat{c}\}}}{r}$, T'_{i,n_1,n_2} is the ith value of T'_{i,n_1,n_2} under H_1^{**}

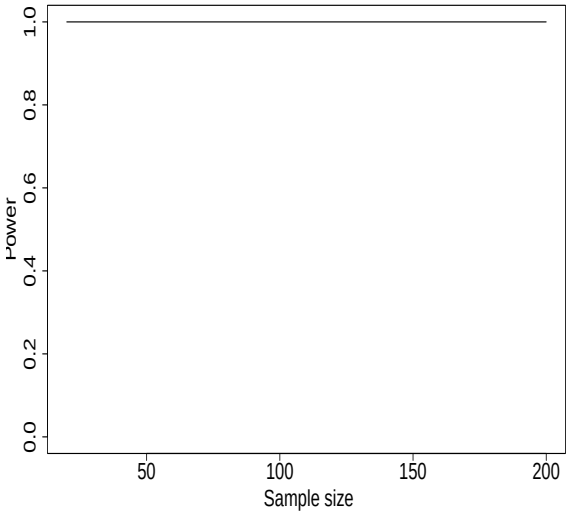
the qth power of the pth Wasserstein distance, where $p = q = 1$ and for the landscape test we have calculated the mean of first landscape function over 1000 grid points.

4.1.1 Circle

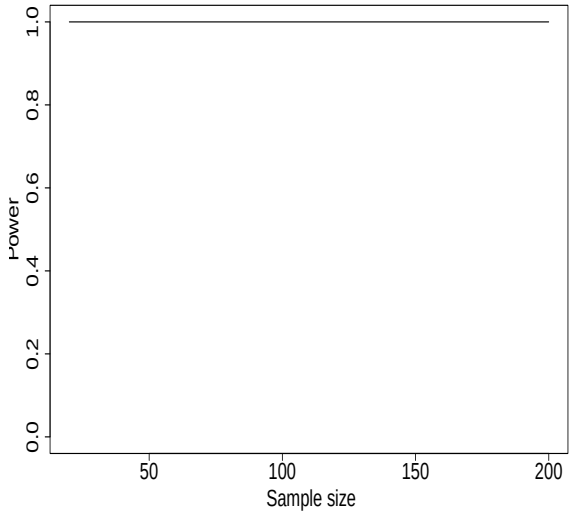
We generate a random sample of size n from the von Mises distribution and apply the Algorithm 1 for $\boldsymbol{\beta_0} = (1,1)^T, \epsilon = n^{-1/d}, r = 10$ and $\alpha = 0.05$. Then we increase the sample size n to examine the consistency of the test. Note that here $d = 2$ for the circle and we take $n \in \{20, 50, 100, 150, 200\}$. The plots in Figure 2 suggest that the test is consistent under the two alternative support unit disk and support of the bivariate normal distribution with mean vector $(0,0)^T$ and covariance matrix $\begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$.

4.1.2 3-D von Mises vs 3-D Multivariate normal

We generate two random samples of the same size n from the 3-D von Mises and 3-D multivariate normal distribution and apply the Algorithm 2 for $\epsilon = n^{-1/d}, r = 10$ and $\alpha = 0.05$. Then we increase the sample size n to examine the consistency of the test. Note that here $d = 3$ and we take $n \in \{20, 50, 100, 150, 200\}$. The plots in Figure 3 suggest that the test is consistent and



(a) Circle vs Disk



(b) Circle vs support of the bivariate normal

Figure 2: Power plots for Circle.

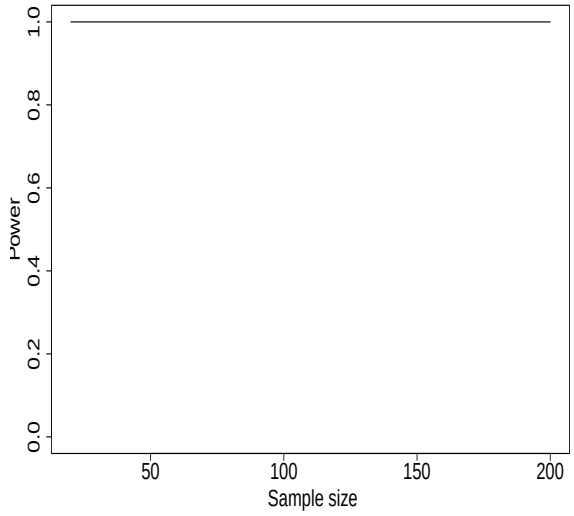
performs better than the other tests.

4.1.3 3-D Sphere vs Unit cube

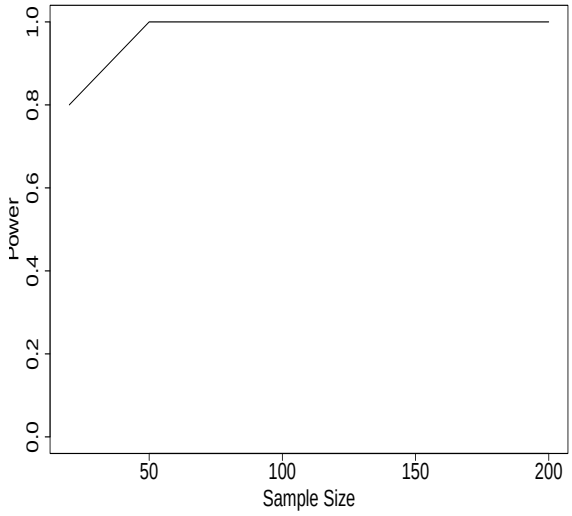
We generate two random samples of the same size n from the 3-D Sphere and unit cube according to uniform distribution and apply the Algorithm 2 for $\epsilon = n^{-1/d}, r = 10$ and $\alpha = 0.05$. Then we increase the sample size n to examine the consistency of the test. Note that here $d = 3$ and we take $n \in \{20, 50, 100, 150, 200\}$. The plots in Figure 4 suggest that the test is consistent and performs equally well in comparison to Robinson's test and landscape test and performs better than the permutation test.

4.2 The supercritical regime

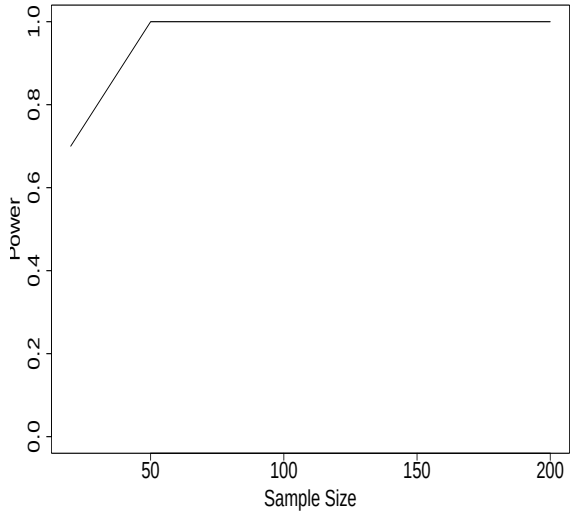
In this subsection, we provide a simulated data study for one and two sample tests under the supercritical regime. We generate a random sample from the unit disk according to the uniform distribution and apply one sample homological equivalence testing procedure. Further,



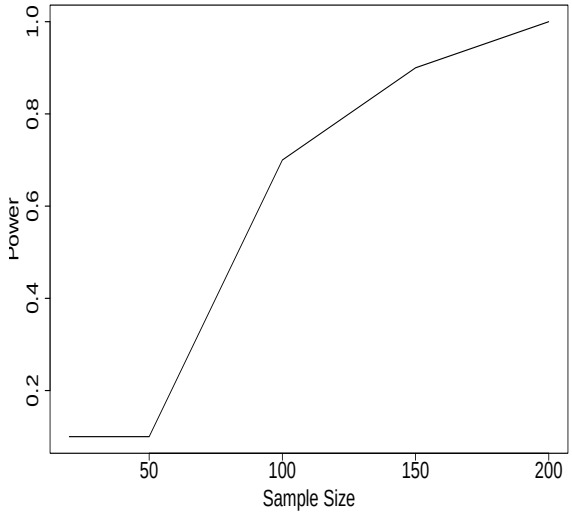
(a) Test based on T_{n_1,n_2}



(b) Robinson's test



(c) Landscape test



(d) Permutation test

Figure 3: von Mises vs Multivariate normal.

we provide a simulation study for two-sample test for 3-D sphere vs support of Swiss roll data set (See, R library tdaunif) and 3-D torus vs 3-D sphere. Moreover, we compare power of two sample tests with Robinson's test, landscape test, and permutation test.